

210B: Lectures 8-10

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Lecture 8

Proposition. Every nonzero prime ideal in a PID is maximal.

Proposition. If R is any commutative ring such that the polynomial ring $R[x]$ is a PID, then R is a field.

Proposition. In an integral domain, a prime element is irreducible.

Lecture 9

Proposition. In a PID, a nonzero element is a prime if and only if it is irreducible.

Proposition. In a unit factorization domain, a non-zero element is prime if and only if it is irreducible.

Lecture 10

Proposition. Let I be an ideal of R and let $(I) = I[x]$ (polynomials with coefficient in the ideal) generated by I in $R[x]$. (polynomials with coefficients in the ideal I) Then

$$R[x]/(I) \cong (R/I)[x].$$

In particular, if I is a prime ideal of R , then (I) is a prime ideal.

Theorem. Let F be a field. Then $F[x]$ is a Euclidean Domain. Specifically, if $a(x), b(x) \in F[x]$ such that $b(x) \neq 0$. Then there exists a unique $q(x), r(x) \in F[x]$ such that

$$a(x) = q(x)b(x) + r(x)$$

where $r(x) = 0$ or $\deg(r(x)) < \deg(b(x))$.